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Garden of Knowledge and Virtue

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Title: “ Chatbot For Automating Restaurant Booking And Ordering Process”

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1.0 ABSTRACT

Chatbots, commonly referred as chatter robots, are computer programs that replicate human conversation through textual or voice communications. One of Chatbot's earliest and perhaps most essential goals was to impersonate a smart person and make it much harder for others to figure out who they really were. We studied in the Natural Language Processing (NLP) domain on this project, which really is a discipline of computer science that uses computational approaches to learn, analyse, and produce human language material. NLP may serve a variety of purposes, including assisting human-human communication (as in machine translation) and human-computer communication (as in conversational bots). Text mining and natural language strategies are commonly employed in customer service applications to anticipate appropriate responses to customers, reducing dependency on call centre operations dramatically. [3]

We made a comprehensive evaluation of relevant research in this work. We examined a range of publications regarding chatbots throughout the past five years. Then we displayed several additional works that were pertinent to our topic, as well as the basic ideas needed to build a smart chatting bot. Consequently, we developed a completely intelligent chatbot to optimize the booking and purchasing process at a restaurant.

Our project's major goal was to create an automatic response system (Chatbot) that answers consumer inquiries on communication networks (Telegram) in order to hasten a restaurant's reservation and purchasing procedure. Using a chatbot, the main objective was to enhance consumer satisfaction and engagement. This technique might be one of the most important axes for obtaining client data in order to carry out marketing tactics in a two-way conversation that aids business sales. The major objective of this project was on constructing a conversational intelligent agent utilising the DialoffFlow framework to automatically generate dialogue "Chat" between a machine and a person. This will give clients with the most up-to-date data and responses from a reliable source at the earliest possible moment.

2.0 INTRODUCTION:

A conversational agent, often referred to as a chatbot, is a software application that can speak with individuals in everyday language. In machine intelligence and natural language processing, dialogue mapping is a major activity. Since its beginnings, building a good chatbot is perhaps the most challenging undertaking in artificial intelligence. Although chatbots may do a broad variety of tasks, their fundamental role is to understand and respond to human language. Previously, basic statistical methodologies or handwriting blueprints and instructions were used to create chatbot designs. As learning capacities have increased, edge neural networks have substituted classical models.

Sophisticated interactive bots can make customers believe they're conversing with a real human, however their ability to expand their expertise instantaneously is limited. The chatbot interprets the data from the user and responds appropriately using artificial intelligence and deep learning techniques. Moreover, they use natural language to connect with humans in a variety of Chat-bot applications, including such as reserving chatbots for restaurants, customer support, and so on. Businesses may find that a chatbot may help them enhance their advertising strategy and increase consumer interest and happiness. Virtual assistants could aid in better management of client data, business management, and online purchasing support, among other things. These are various scenarios in which chatbots might assist medical practitioners by reducing their workload.

2.1 BACKGROUND:

Social media networks are prominent and popular technologies that assist businesses organisations to improve their online trading [3] and customer relations [4], although with culture's widespread acceptance of digitization, better technical solutions at the management level are considered necessary. A chatbot, for example, appears to become a doorway to the automation of many functions formerly handled by human operators or businesses, saving time and money and increased concentrations of user and customer satisfaction [5]. Chatbots are

software programs that participate in a series of conversations to achieve a specific objective and act as a facilitator for the people with whom they interact. Wallace described ALICE chatbot in his review as, "expert systems that use case-based reasoning" (CBR).[6]

Those programs use text messages, consoles, or audio chat to simulate an intelligent conversation with a human counterpart.

This project explains the chatbot creation and incorporation processes in a commercial (restaurant) setting, with a focus on how that might be incorporated into an industry's content marketing techniques early in the sales pipeline. The main objective was to employ a Telegram chatbot powered by the DialogFlow platform to boost the amount of revenue earned. A business that advertises itself digitally may use the DialogFlow platform to create a chatbot to help them gain more customers.

2.2 OBJECTIVE:

The overall goal was to investigate how chatbots in online business may boost customer engagement. The following were the precise goals:

1. Create a "Table reservation and Ordering" chatbot for a restaurant .
2. Experiencing various NLP techniques for conversational agents.
3. Work with Google DialogFlow API for bot implementation.
4. Examine the platform's usability (installation, usage, data processing, and efficacy).
5. Integrate the bot with Telegram messaging platform
6. Examine the outcomes of the implementation.

3.0 LITERATURE REVIEW

For some decades, AI experts have been dedicated to developing computational dialogue models (chatbots). Many fields, especially philosophy, linguistics, computer science, and sociology, are used to create complex AI conversational and dialogue platforms [5]. This subsection will look at historical chatbot research and implementations.

3.1 Chatbots Applications and Uses:

Chatbots are dynamic chatting programs that use artificial conversation algorithms. Chatbots have been around for a long period of time; Joseph Weizenbaum's Eliza software, built in the early 1960s, was the very first well-known chatbot. Eliza enhanced human-machine interaction by simulating a discussion using minimal pattern recognition as well as a pattern response mechanism [6, 7].

Chatbots have become essential in many aspects of life, with one of the most common usage in academia being a question-answering software for a certain body of knowledge. The authors suggested a methodology that was built as a virtual assistant to help students learn the Java programming language in [8]. The built model has been tested to see how people react to the system's engagement. In addition, as described in [9], a chatbot specialized in student management difficulties might assist students in enrolling for and abandoning courses. The administrative student chatbot assists colleges with automated inquiry responses 24 hours a day, seven days a week, and ensures that students receive reliable information from a reliable source.

The employment of chatbot technology in organisations to answer customer's issues and aid the marketing strategy is referred to as customer service agent. Companies are going to the internet world to improve the consumer experience in purchase and give advanced technological assistance ways of handling after-sales difficulties as organisations become more international throughout the new age of internet advertising and artificial intelligence. E-service agents are also becoming more popular as an alternative to fashion businesses like Burberry, Louis Vuitton, Tommy Hilfiger, Levi's, H&M, and eBay [1].

The healthcare bot will operate as if it were a simulated doctor, dealing with patients. Pattern recognition methods and Natural Language Processing techniques are used to create this chatbot, which is written in Python. As shown in a poll done to assess the chatbot's effectiveness, the chatbot provided 80 percent right responses, while the chatbot provided 20 percent ambiguous/incorrect replies. This chatbot may predominantly is being used as a digital physician for healthcare and consciousness, as well as for training medical students, according to the findings of this study.[5]

Artificial Intelligence could be used to anticipate any ailment and present a list of possible treatments depending on the indications provided. Furthermore, if a person's body is analysed on a regular basis, AI can assist anticipate any potential sickness well before damage has been done to the person. The research and development costs, as well as state backing for the correct and effective deployment of all drugs, are key issues for this subject that were not discussed in this research article. [8]

4.0 METHODOLOGY:

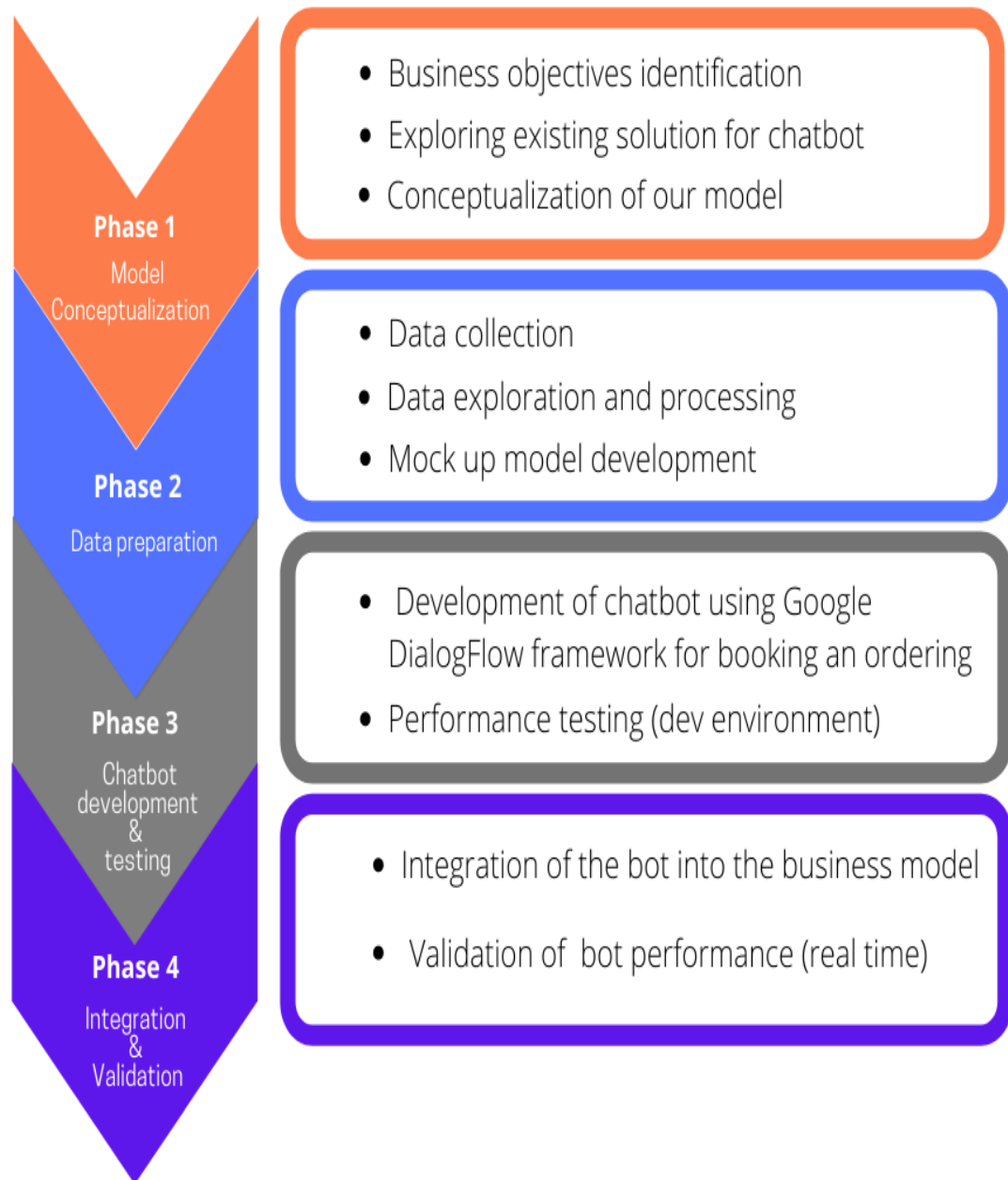


Fig-1 Chatbot Development Methodology

4.1 FLOW CHART:

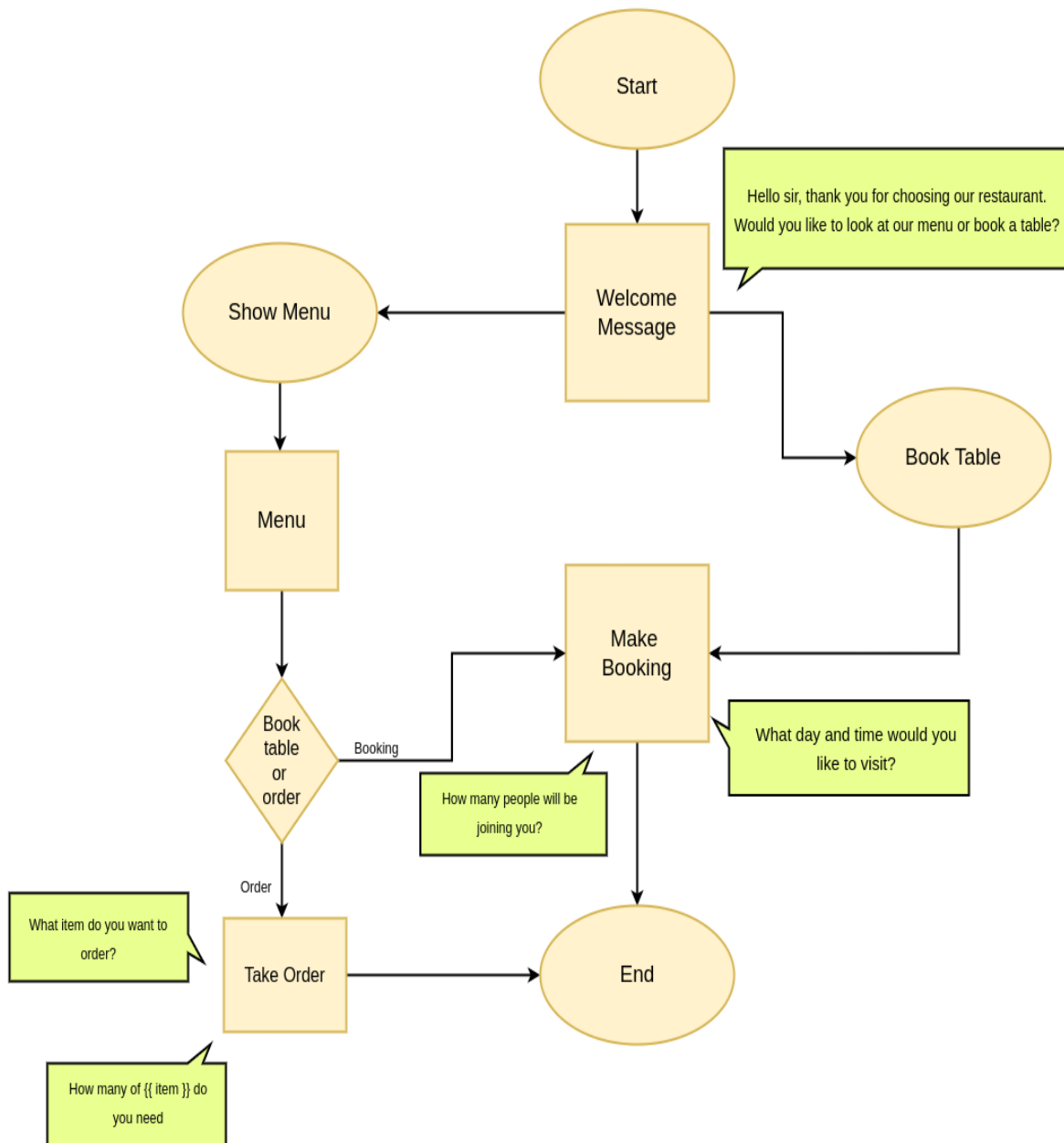


Fig-2 Flow Chart of ChatBot

4.2 PSEUDOCODE:

1. **Initiate** ChatBot
2. **Print** Welcome Message
3. **HandleUserInput**(message):
4. **Function** HandleUserInput(message):
 - a. $processed_message \leftarrow \text{ProcessMessage}(message)$
 - b. $message_intent \leftarrow \text{GenerateIntentFromMessage}(processed_message)$
 - c. **If** $message_intent == \text{"show menu"}$:
 - i. **Print** Menu
 - d. **Else if** $message_intent == \text{"booking"}$:
 - i. **ReserveTable**(data)
 - e. **Else if** $message_intent == \text{"order"}$:
 - i. **CreateOrder**(data)
 - f. **Else if** $message_intent == \text{"good_bye"}$:
 - i. **Print** Goodbye message
 - ii. **return**
 - g. **HandleUserInput**(message)
5. **Terminate** ChatBot

5.0 RESULTS:

The system's output is shown in the following figures.

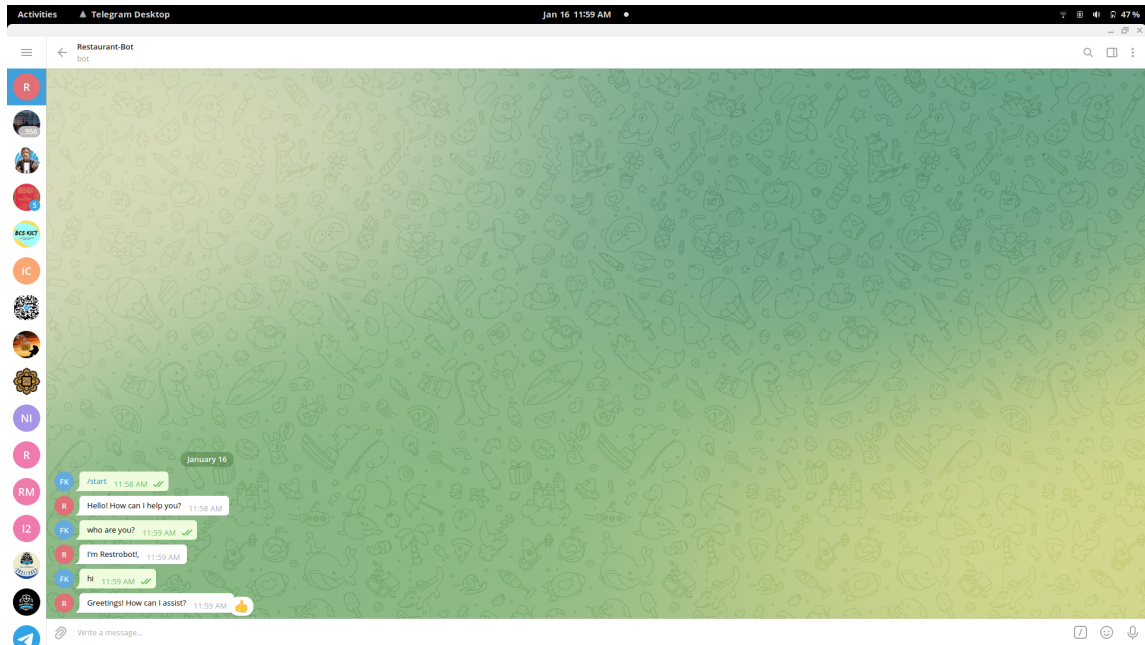


Fig-3: Greetings from the Bot

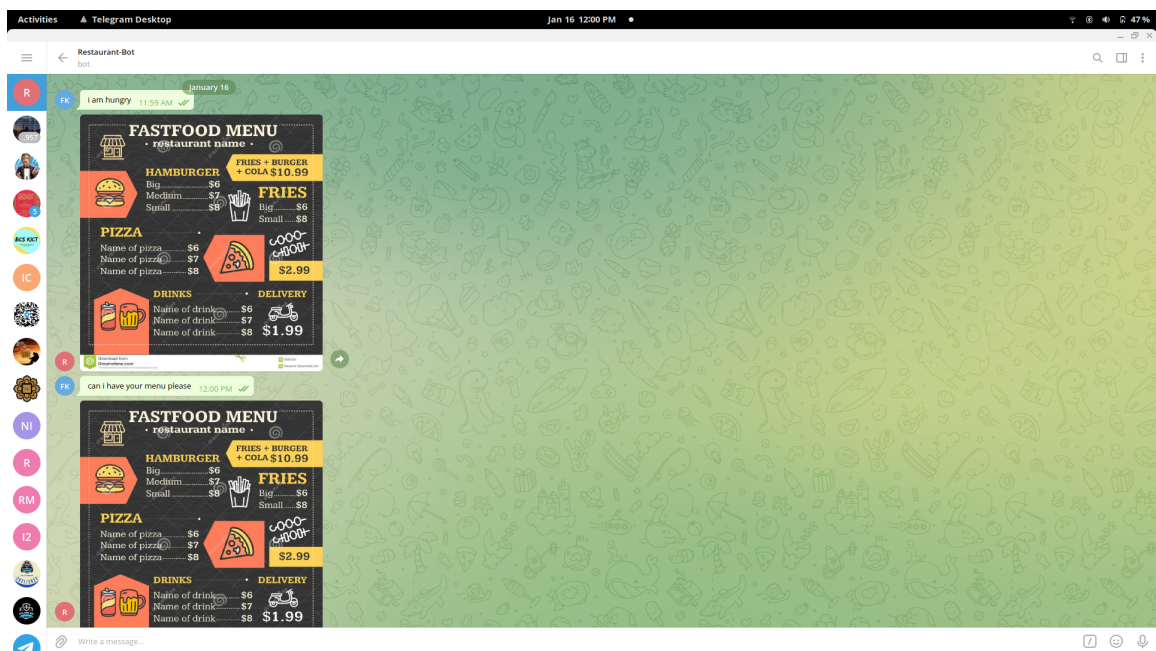


Fig-4: Bot response for show menu query

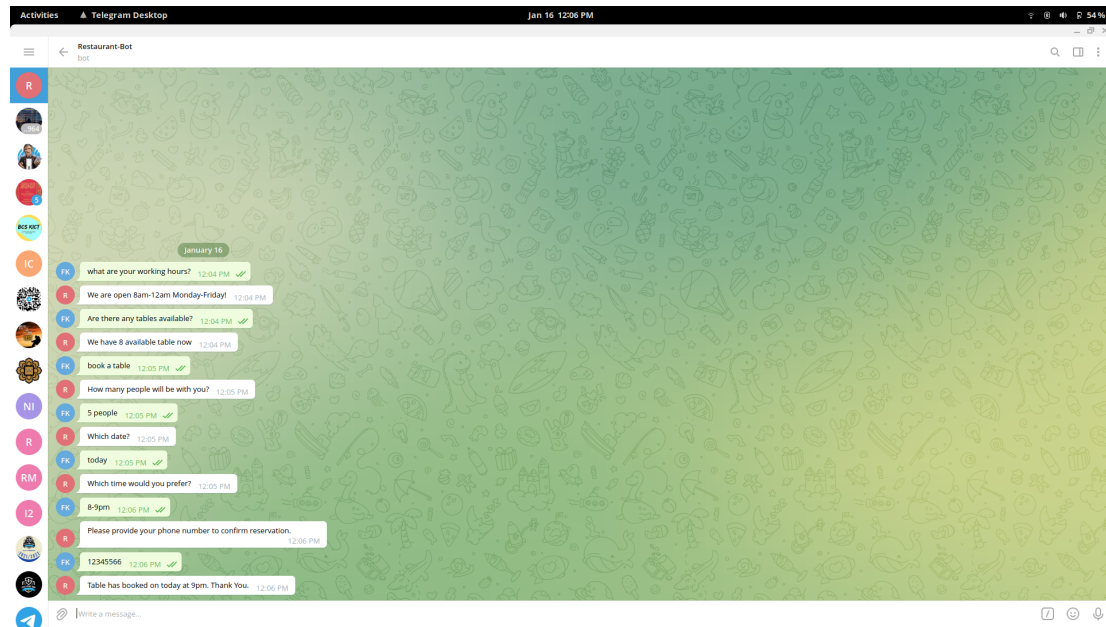


Fig-5: Table Reservation

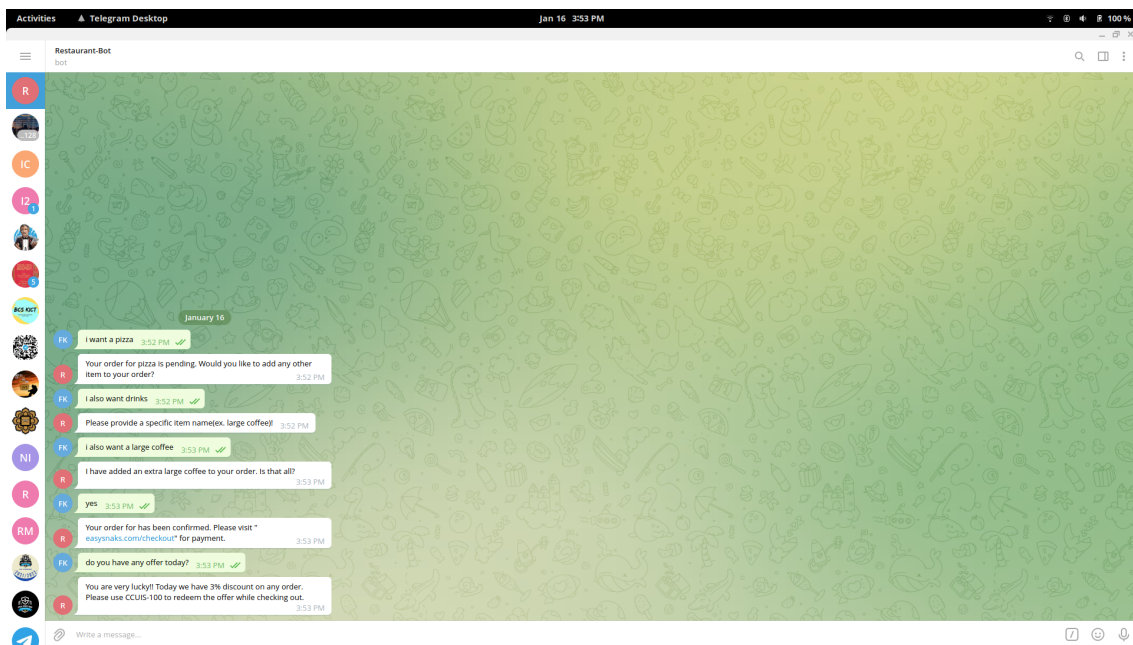


Fig-6: Order Completion and Coupon generation



Fig-7: Bot response for general query(address, contact, operating hour)



Fig-8 : Goodbye message from the Bot

6.0 CONCLUSION

This project presents a chatbot for automating a restaurant's ordering and booking procedure. The usage of the DialogFlow software for Telegram Messenger to support the restaurant's business model is an important instrument for carrying out duties linked to ordering and booking mechanisms, which saves money and enhances the customer experience. We provide features like warm greetings and clear inquiries to influence customer behaviour and elicit personal information; for example, consumers may enter their email address through the chatbot and become prospective leads for the organisation.

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